ALKADIYENLAR

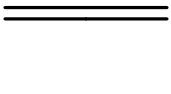
Diyen uglevodorodlar · umumiy formula C_nH_{2n-2}

Student book – nazariy qo'llanma

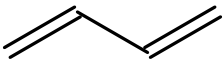
Uglerod zanjirida **ikkita qo'sh bog'** bo'lgan to'yinmagan uglevodorodlar **alkadiyenlar** (diyen uglevodorodlar) deyiladi. Umumiy formulasi $\mathbf{C_nH_{2n-2}}$ ($n \geq 3$). Ular **alkinlar** bilan sinflararo izomer.

Qo'sh bog'larning joylashishiga ko'ra turlari

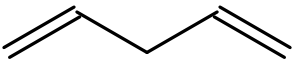
- **Kumulyativ** (allen tipidagi) – qo'sh bog'lar bitta uglerodda yonma-yon: $CH_2=C=CH_2$.
- **Konyugirlangan** (tutash, butadiyen-1,3 tipidagi) – qo'sh va oddiy bog'lar navbatma-navbat: $CH_2=CH–CH=CH_2$.
- **Izolyatsiyalangan** (ajralgan) – qo'sh bog'lar ikki va undan ortiq oddiy bog' bilan ajralgan: $CH_2=CH–CH_2–CH=CH_2$.



kumulyativ: allen



konyugirlangan: butadiyen-1,3



izolyatsiyalangan: pentadiyen-1,4

Tuzilishi

- **Butadiyen-1,3:** to'rtala uglerod sp^2-gibridlangan, molekula yassi. To'rtta p-orbital o'zaro qoplanib **yagona π-elektron tizimi** hosil qiladi – bu **konyugatsiya** (tutashuv) deyiladi.
- Konyugatsiya natijasida bog' uzunliklari tenglashadi: $C_1=C_2 \approx 0{,}137$ nm (oddiy qo'sh bog' 0,134 nm), $C_2–C_3 \approx 0{,}147$ nm (oddiy bog' 0,154 nm). Shuning uchun konyugirlangan diyenlar barqarorroq va ularga **1,4-birikish** xos.
- **Allen:** o'rtadagi uglerod sp, chetdagilari sp^2-gibridlangan; C=C=C chiziqli (180°), ikkala π-bog' o'zaro perpendikulyar tekisliklarda yotadi.
- Izolyatsiyalangan diyenlarda har bir qo'sh bog' alohida, xossalari oddiy alkenlarnikiga o'xshaydi.

Umumiy formulalar (C_nH_{2n-2})

Ko'rsatkich	Ifoda	Misol: butadiyen-1,3 C_4H_6
Molyar massa	$M_r = 14n - 2$	54
Vodorod bo'yicha zichlik	$D(H_2) = 7n - 1$	27
σ-bog'lar soni	$3n - 3$	9
π-bog'lar soni	2	2
C–H bog'lari	$2n - 2$	6
C–C bog'lari (σ)	$n - 1$	3
Jami atomlar	$3n - 2$	10
Yonishi	$C_nH_{2n-2} + (1{,}5n - 0{,}5)O_2 \rightarrow nCO_2 + (n-1)H_2O$	$2C_4H_6 + 11O_2 \rightarrow 8CO_2 + 6H_2O$

1. Nomlanishi

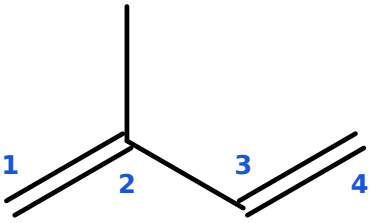
A) Sistematik nomenklatura

- **Ikkala qo'sh bog'** saqlagan eng uzun zanjir tanlanadi.
- Zanjir **qo'sh bog'lar yaqin** tomondan raqamlanadi (qo'sh bog'lar raqami eng kichik bo'lishi kerak).
- Radikallar **alifbo tartibida**, o'rni ko'rsatilgan holda aytiladi.
- Alkan nomidagi «-an» o'rniga **«-adiyen»** qo'shiladi va ikkala qo'sh bog' o'rni yoziladi: butan → **butadiyen-1,3**; pentan → pentadiyen-1,4.

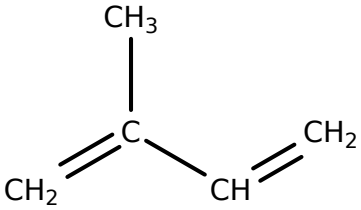
Raqamlashga misollar (chapda skelet, o'ngda yoyilgan formula)

a) 2-metilbutadiyen-1,3 (izopren)

skelet formula

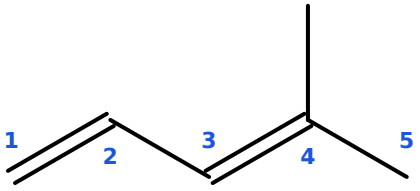


yoyilgan (to'liq) formula

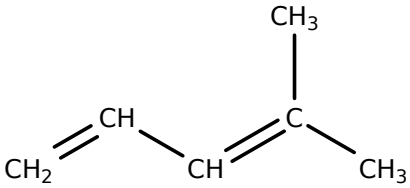


b) 4-metilpentadiyen-1,3 (qo'sh bog'lar chap tomonga yaqin)

skelet formula

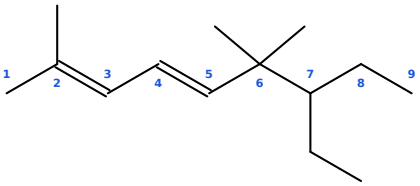


yoyilgan (to'liq) formula

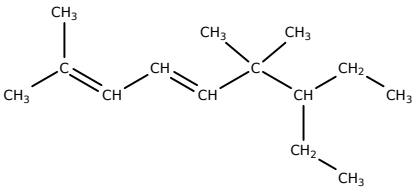


c) 7-etil-2,6,6-trimetilnonadiyen-2,4

skelet formula



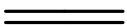
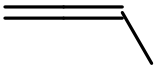
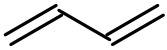
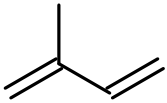
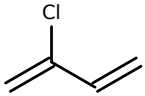
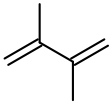
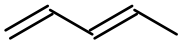
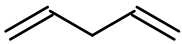
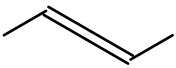
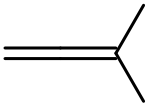
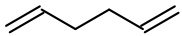
yoyilgan (to'liq) formula

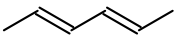
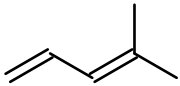


B) Tarixiy (trivial) nomlar

Ko'p uchraydigan diyenlar tarixiy nomlari bilan ataladi: **allen**, **divinil** (butadiyen-1,3), **izopren**, **xloropren**, **piperilen** (pentadiyen-1,3), **diallil** (geksadiyen-1,5).

Nomlash misollari (to'plam jadval)

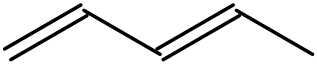
Oddiy formula	Skelet formula	Sistematik	Tarixiy
$CH_2=C=CH_2$		propadiyen	allen
$CH_2=C=CH–CH_3$		butadiyen-1,2	metilallen
$CH_2=CH–CH=CH_2$		butadiyen-1,3	divinil
$CH_2=C(CH_3)–CH=CH_2$		2-metilbutadiyen-1,3	izopren
$CH_2=C(Cl)–CH=CH_2$		2-xlorbutadiyen-1,3	xloropren
$CH_2=C(CH_3)–C(CH_3)=CH_2$		2,3-dimetilbutadiyen-1,3	metilizopren
$CH_2=CH–CH=CH–CH_3$		pentadiyen-1,3	piperilen
$CH_2=CH–CH_2–CH=CH_2$		pentadiyen-1,4	divinilmetan
$CH_3–CH=C=CH–CH_3$		pentadiyen-2,3	dimetilallen
$CH_2=C=C(CH_3)–CH_3$		3-metilbutadiyen-1,2	–
$CH_2=CH–CH_2–CH_2–CH=CH_2$		geksadiyen-1,5	diallil

Oddiy formula	Skelet formula	Sistematik	Tarixiy
$CH_3–CH=CH–CH=CH–CH_3$		geksadiyen-2,4	–
$CH_2=CH–CH=C(CH_3)–CH_3$		4-metilpentadiyen-1,3	–

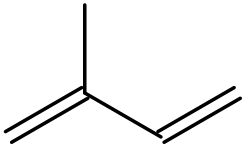
2. Izomeriyasi

Alkadiyenlarda: uglerod **zanjiri**, qo'sh bog'lar **holati**, **sinflararo** va **fazoviy** (sis-trans) izomeriya uchraydi.

1) Uglerod zanjiri izomeriyasi (C_5H_8)

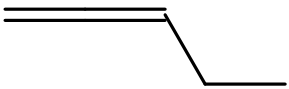


pentadiyen-1,3

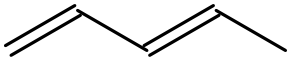


2-metilbutadiyen-1,3 (izopren)

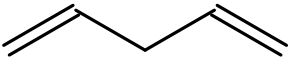
2) Qo'sh bog'lar holati izomeriyasi (C_5H_8)



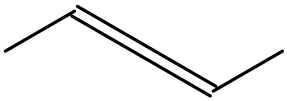
pentadiyen-1,2



pentadiyen-1,3



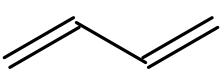
pentadiyen-1,4



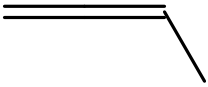
pentadiyen-2,3

3) Sinflararo izomeriya

Alkadiyenlar $C_nH_{2n–2}$ umumiy formulali **alkinlar** (hamda sikloalkenlar) bilan izomer. C_4H_6 uchun misollar:



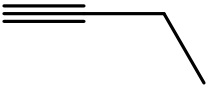
butadiyen-1,3



butadiyen-1,2



siklobuten



butin-1



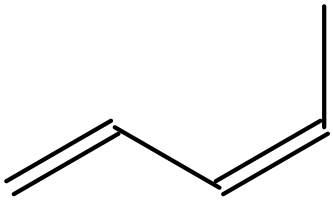
butin-2

4) Fazoviy (sis-trans) izomeriya

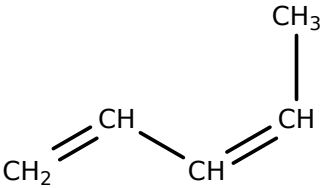
Qo'sh bog'dagi har bir uglerod ikki xil o'rinbosarga ega bo'lsa, sis-trans izomeriya mumkin. Masalan, **pentadiyen-1,3** (piperilen) da $C_3{=}C_4$ bog' bo'yicha:

sis-pentadiyen-1,3

skelet formula

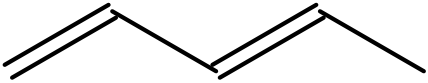


yoyilgan (to'liq) formula

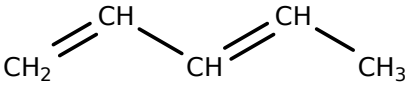


trans-pentadiyen-1,3

skelet formula

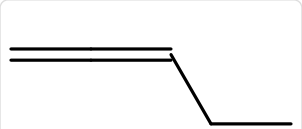


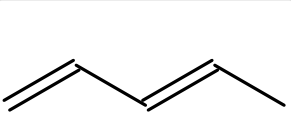
yoyilgan (to'liq) formula

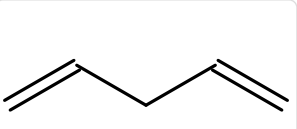


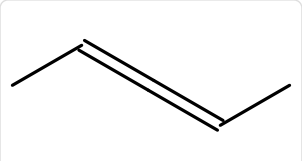
Butadiyen-1,3, izopren va allen sis-trans izomerga ega emas. Geksadiyen-2,4 da ikkala qo'sh bog' bo'yicha uchta fazoviy izomer bor: trans,trans; sis,sis; sis,trans.

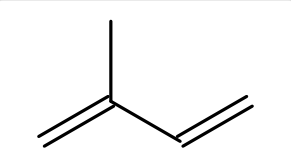
C_5H_8 alkadiyenlari (6 ta struktura izomeri)

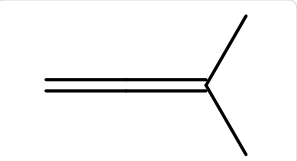

pentadiyen-1,2


pentadiyen-1,3


pentadiyen-1,4


pentadiyen-2,3


2-metilbutadiyen-1,3


3-metilbutadiyen-1,2

Pentadiyen-1,3 ning sis- va trans-shakllarini hisobga olsak – 7 ta. C_5H_8 tarkibli alkinlar: pentin-1, pentin-2, 3-metilbutin-1 (3 ta).

3. Fizik xossalari

- Allen va **butadiyen-1,3** – normal sharoitda **gaz**; izopren, xloropren, 2,3-dimetilbutadiyen-1,3 va boshqa diyenlar – **suyuqlik**.
- Suvda deyarli erimaydi, organik erituvchilarda yaxshi eriydi; zichligi suvnikidan kichik.
- M_r ortishi bilan t_q va zichlik ortadi. Butadiyen-1,3 – o'ziga xos hidli rangsiz gaz.

Modda	Formula	Holati	t_q, °C
allen	C_3H_4	gaz	–34
butadiyen-1,3	C_4H_6	gaz	–4,4
pentadiyen-1,4	C_5H_8	suyuq	26
izopren	C_5H_8	suyuq	34
piperilen (trans)	C_5H_8	suyuq	42
xloropren	C_4H_5Cl	suyuq	59
2,3-dimetilbutadiyen-1,3	C_6H_{10}	suyuq	69

4. Olinishi

1) Lebedev usuli – etil spirtdan (MgO, ZnO yoki Al_2O_3, 400–500 °C)

$$2CH_3–CH_2–OH \xrightarrow{ZnO,\ 450\ °C} CH_2=CH–CH=CH_2 + 2H_2O + H_2$$

S.V. Lebedev 1928-yilda taklif qilgan; 1932-yildan butadiyen sanoatda shu usulda olina boshlagan.

2) Alkanlar va alkenlarni degidrogenlash (Cr_2O_3/Al_2O_3, ≈ 600 °C)

$$CH_3–CH_2–CH_2–CH_3 \xrightarrow{t,\ kat.} CH_2=CH–CH=CH_2 + 2H_2$$

$$CH_2=CH–CH_2–CH_3 \xrightarrow{t,\ kat.} CH_2=CH–CH=CH_2 + H_2$$

$$CH_3–CH(CH_3)–CH_2–CH_3 \xrightarrow{t,\ kat.} CH_2=C(CH_3)–CH=CH_2 + 2H_2$$

(izopentan → izopren)

3) Glikollarni degidratlash

$$CH_2(OH)–CH_2–CH(OH)–CH_3 \xrightarrow{H+,\ t} CH_2=CH–CH=CH_2 + 2H_2O$$

4) Digalogenalkanlarga KOH ning spirtli eritmasi ta'siri

$$CH_2Br–CH_2–CH_2–CH_2Br + 2KOH \xrightarrow{spirt,\ t} CH_2=CH–CH=CH_2 + 2KBr + 2H_2O$$

5) Xloropren – vinilatsetilenga HCl biriktirib

$$CH{\equiv}C–CH=CH_2 + HCl \xrightarrow{kat.} CH_2=C(Cl)–CH=CH_2$$

6) Allen – 2,3-dibrompropenga rux ta'siri

$$CH_2Br–CBr=CH_2 + Zn \longrightarrow CH_2=C=CH_2 + ZnBr_2$$

5. Kimyoviy xossalari

Ikkita π-bog' borligi uchun diyenlar alkenlarga o'xshab **birikish**, **oksidlanish** va **polimerlanish** reaksiyalariga kirishadi. Konyugirlangan diyenlar uchun **1,2- va 1,4-birikish** xos: mahsulot diyenning tuzilishi, reaksiya sharoiti, erituvchi va reagentga bog'liq.

1,2-birikish – reagent bitta qo'sh bog'ga (C_1 va C_2 ga) birikadi. **1,4-birikish** – reagent chetki C_1 va C_4 ga birikadi, qo'sh bog' o'rtaga (C_2=C_3) ko'chadi. Past haroratda 1,2-, yuqoriroq haroratda asosan 1,4-mahsulot hosil bo'ladi.

I. Birikish reaksiyalari

1) Gidrogenlash (Ni, Pt, t)

$$CH_2{=}CH{-}CH{=}CH_2 + H_2 \xrightarrow{1,2} CH_3{-}CH_2{-}CH{=}CH_2 \quad \text{buten-1}$$

$$CH_2{=}CH{-}CH{=}CH_2 + H_2 \xrightarrow{1,4} CH_3{-}CH{=}CH{-}CH_3 \quad \text{buten-2}$$

$$CH_2{=}CH{-}CH{=}CH_2 + 2H_2 \xrightarrow{Ni,\ t} CH_3{-}CH_2{-}CH_2{-}CH_3 \quad \text{butan}$$

2) Galogenlash (I. Tile o'rgangan)

$$CH_2{=}CH{-}CH{=}CH_2 + Br_2 \xrightarrow{1,2} CH_2Br{-}CHBr{-}CH{=}CH_2$$

3,4-dibrombuten-1

$$CH_2{=}CH{-}CH{=}CH_2 + Br_2 \xrightarrow{1,4} CH_2Br{-}CH{=}CH{-}CH_2Br$$

1,4-dibrombuten-2

$$CH_2{=}CH{-}CH{=}CH_2 + 2Br_2 \longrightarrow CH_2Br{-}CHBr{-}CHBr{-}CH_2Br$$

brom mo'l bo'lsa: 1,2,3,4-tetrabrombutan. Bromli suv rangsizlanadi – sifat reaksiya.

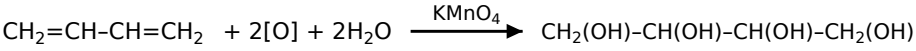
3) Galogenvodorodlarning birikishi

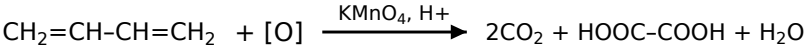
$$CH_2{=}CH{-}CH{=}CH_2 + HCl \xrightarrow{1,2} CH_3{-}CHCl{-}CH{=}CH_2 \quad \text{3-xlorbuten-1}$$

$$CH_2{=}CH{-}CH{=}CH_2 + HCl \xrightarrow{1,4} CH_3{-}CH{=}CH{-}CH_2Cl \quad \text{1-xlorbuten-2}$$

II. Oksidlanish

$KMnO_4$ eritmasini **rangsizlantiradi** (sifat reaksiya). Yumshoq oksidlanganda (Vagner) to'rt atomli spirt, qattiq oksidlanganda qo'sh bog'lar uzilib kislotalar va CO_2 hosil bo'ladi:

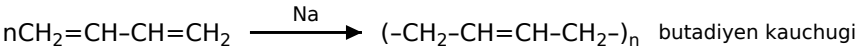


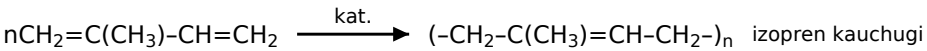


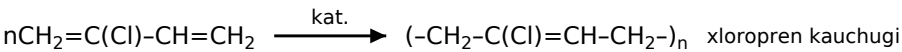
Yonishi: $2C_4H_6 + 11O_2 \rightarrow 8CO_2 + 6H_2O$

III. Polimerlanish va sopolimerlanish

Diyenlar katalizator (masalan, Na) yoki ultrabinafsha nur ta'sirida asosan **1,4-holat** bo'yicha, radikal mexanizmda oson polimerlanadi; har bir zvenoda bitta qo'sh bog' qoladi. Zvenolar sis- yoki trans-konfiguratsiyada bo'lishi mumkin.







Sopolimerlanish – ikki xil monomerning birgalikda polimerlanishi. Butadiyen + stirol (C_6H_5–CH=CH_2) → **butadiyen-stirol kauchugi**; butadiyen + akrilonitril (CH_2=CH–CN) → **butadiyen-nitril kauchugi** (moy va benzinga chidamli).

6. Kauchuklar va rezina

Tabiiy kauchuk

- Kauchukli o'simliklar (masalan, Braziliya **geveyasi**) sutsimon shirasi – **lateks**dan olinadi.
- Tarkibi **$(C_5H_8)_n$** – izopren polimeri; tuzilishi **sis-1,4-poliizopren**. Trans-1,4-poliizopren – **guttapercha** (elastik emas).
- Elastik, suv va gaz o'tkazmaydi, elektr tokini o'tkazmaydi; benzinda bo'kadi va eriydi. Issiqda yopishqoq, sovuqda mo'rt bo'lib qoladi.

Vulkanizatsiya

Kauchukni **oltingugurt** bilan qizdirish (Ch. Gudyir, 1839). Makromolekulalar orasida –S–S– «ko'priklar» hosil bo'lib, to'rsimon tuzilish yuzaga keladi: mustahkamlik, elastiklik va haroratga chidamlilik ortadi.

Mahsulot	Oltingugurt miqdori	Xossasi
rezina	≈ 1–3 %	elastik, mustahkam
ebonit	≈ 30–40 %	qattiq, elektr izolyatori

Sintetik kauchuklar

Kauchuk	Monomer	Xususiyati, ishlatilishi
butadiyen (SKB)	butadiyen-1,3	birinchi sintetik kauchuk (Lebedev); shina, rezina buyumlar
izopren (SKI)	izopren	tabiiy kauchukka eng yaqin; shina, tibbiy buyumlar
xloropren (neopren)	xloropren	moy, benzin va olovga chidamli; shlang, kabel qobig'i, yelim
butadiyen-stirol (SKS)	butadiyen + stirol	yeyilishga chidamli; avtoshina, poyabzal tagi
butadiyen-nitril (SKN)	butadiyen + akrilonitril	moy-benzinga chidamli; zichlagichlar, shlanglar

7. Ishlatilishi

- **Butadiyen-1,3** – sintetik kauchuklar (SKB, SKS, SKN) va plastmassalar (masalan, ABS-plastik) ishlab chiqarishning asosiy xomashyosi.
- **Izopren** – izopren kauchugi va butil kauchuk (izobutilen bilan) olishda.
- **Xloropren** – moy-benzinga chidamli neopren kauchugi olishda.
- **Rezina buyumlar:** avtomobil shinalari, konveyer tasmalari, shlanglar, poyabzal, tibbiy qo'lqop va buyumlar, sport anjomlari.
- **Ebonit** – elektr izolyatsion materiallar.

Diyenlarni aniqlash: bromli suv va $KMnO_4$ eritmasi rangsizlanadi (alkenlar kabi). 1 mol konyugirlangan diyen 2 mol gacha Br_2 yoki H_2 biriktiradi.

O'zingni tekshir

1. C_5H_8 tarkibli alkadiyenlarning barcha struktura izomerlarini yozing va nomlang.

2. Izoprenning sistematik nomini yozing.

3. Butadiyen-1,3 ga 1 mol Br_2 birikkanda qanday mahsulotlar hosil bo'ladi?

4. Lebedev usuli tenglamasini yozing. 92 g etanoldan necha gramm butadiyen olish mumkin (nazariy)?

5. 5,4 g butadiyen-1,3 bilan necha gramm brom to'liq birikadi?

6. Tabiiy kauchukning tarkibi va tuzilishi qanday?

7. Rezina ebonitdan nima bilan farq qiladi?

Javoblar: 1) 6 ta: pentadiyen-1,2; pentadiyen-1,3; pentadiyen-1,4; pentadiyen-2,3; 2-metilbutadiyen-1,3; 3-metilbutadiyen-1,2; 2) 2-metilbutadiyen-1,3; 3) 3,4-dibrombuten-1 (1,2-birikish) va 1,4-dibrombuten-2 (1,4-birikish); 4) $2C_2H_5OH \rightarrow C_4H_6 + 2H_2O + H_2$; 2 mol etanol → 1 mol, 54 g; 5) 0,1 mol $C_4H_6 \times 2 = 0{,}2$ mol $Br_2 = 32$ g; 6) $(C_5H_8)_n$, sis-1,4-poliizopren; 7) oltingugurt miqdori: rezinada ≈ 1–3 %, ebonitda ≈ 30–40 %.